Profiling Academic Trajectories

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DSCI470 Project 2

Introduction & Data Exploration

- Questionnaire responses from 145 UC Irvine Students
- 31 total questions
  - 10 personal questions
  - 6 family questions
  - 15 educational habits questions
- Integer-encoded features
  - Nominal & Ordinal Values

Higher Education Students Performance Evaluation

Donated on 8/14/2023

Figure 1: Chosen Dataset

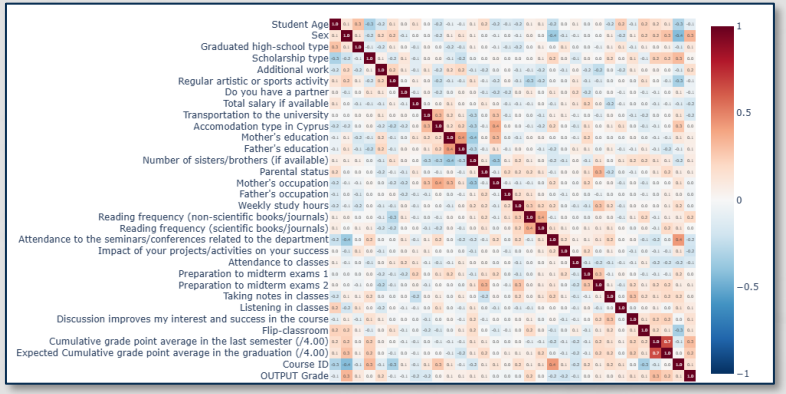

Figure 2: Pearson Correlation Matrix of all features

High School Type	1, 2, 3	Type of high school attended: 1 = private; 2 = public/state; 3 = other.
Scholarship Type	1–5	Financial aid level: 1 = none; 2 = 25%; 3 = 50%; 4 = 75%; 5 = full scholarship.

Figure 3: Example of nominal vs. ordinal features

Research Questions

#1

Student Groups

Do any student clusters emerge from this data?

#2

“Model” Student

What would the “perfect student” look like?

#3

Social Indicators for Attendance

Which social indicators most impact class attendance?

#1) Investigating Student Groups

- DBSCAN Clustering
  - $\varepsilon = 9.0$
- K-Means Clustering
  - Silhouette Analysis
  - Optimal $k = 2$
- Describing Cluster Differences
  - Compared cluster means
  - Mann-Whitney U tests

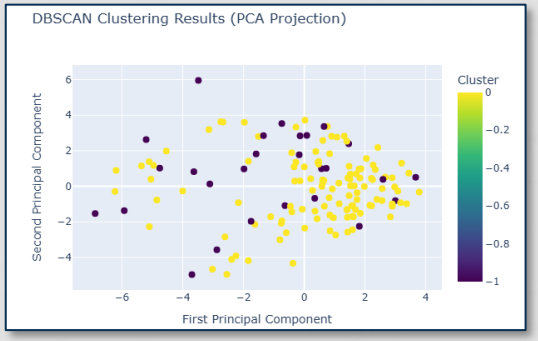

Figure 1: Initial DBSCAN Results

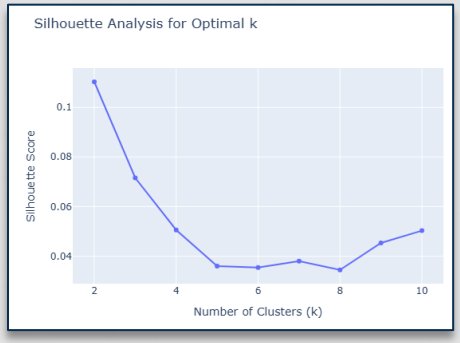

Figure 2: Silhouette Analysis Results

Attribute	Group 1	Group 2
Gender	Mostly female	Mostly male
Engagement	Higher	Lower
Courses	Course 1	Courses 8 & 9
Study Habits	More independent & active	Less independent & active
Learning Perception	Positive impact	Weaker impact
Performance	Higher GPA	Lower GPA

Figure 4: Mann-Whitney U Results Interpretation

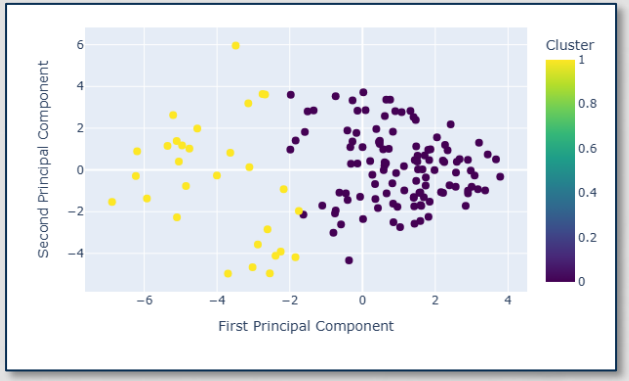

Figure 3: K-Means Clustering Results

#2) Modeling the “Perfect” Student

- Poor Random Forest Classification performance on raw data
- Identified three key limitations
  - Large number of target classes
  - Small dataset with unbalanced classes
  - Complex dataset with noise
- Binned values based on low, mid, and high performers
- Randomized Cross-Validation Search using binned values
  - 65% accuracy
- Extracted initial feature importances

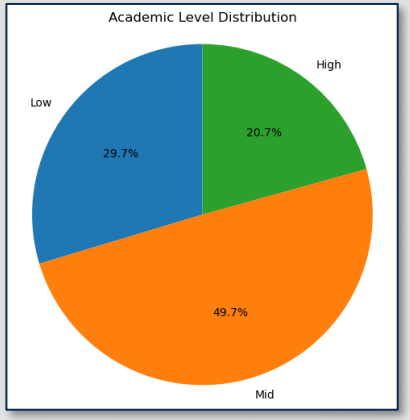

Figure 1: Class Distributions after Binning

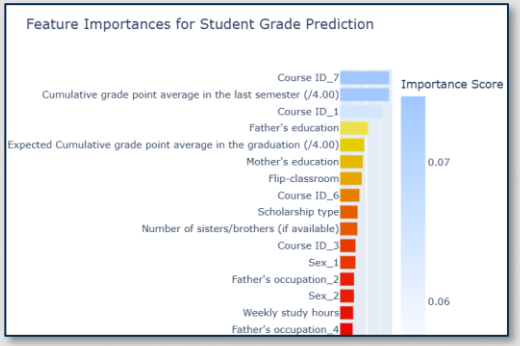

Figure 2: Initial Extracted Feature Importances

#2) Modeling the “Perfect” Student Cont.

- Dropped dominant, non-informative indicators
- Re-ran Randomized Cross-Validation
  - 56% accuracy
- Permutation Importance Analysis
  - Pulled out top 10 results
- SHAP Analysis
  - SHAP dependence plots for each feature

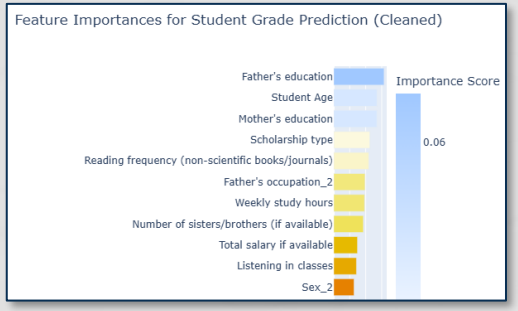

Figure 1: Feature Importances after Dropping Features

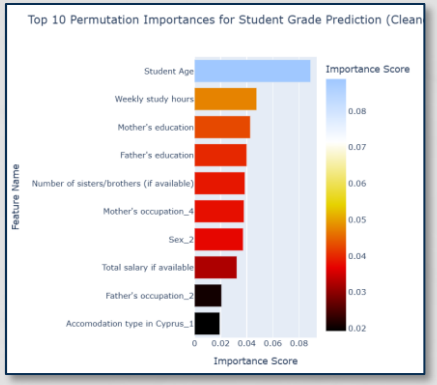

Figure 2: Top 10 Permutation Importances

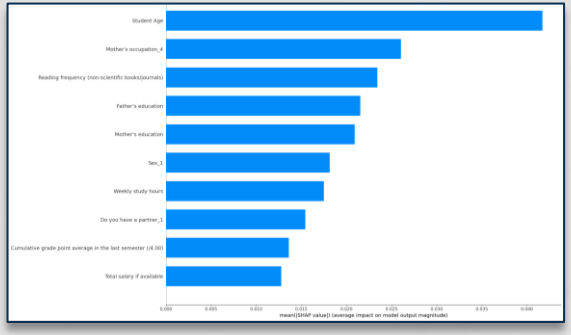

Figure 3: Mean SHAP Values (average impact on model output magnitude)

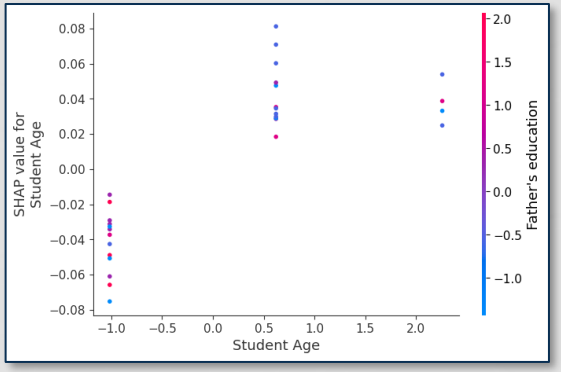

Figure 4: SHAP Dependence Plot for Student Age

#2) Modeling the “Perfect” Student Cont.

- Interpreted SHAP dependence results to reach following conclusions:

Commonalities Among Exceptionally High Performing Students
Older (22-25)
Very *low* or very *high* **Study Hours**
Parents *not* in Private Sector
No or *few* **Siblings**
Lower **Weekly Salary**
Mother with *higher* **education**

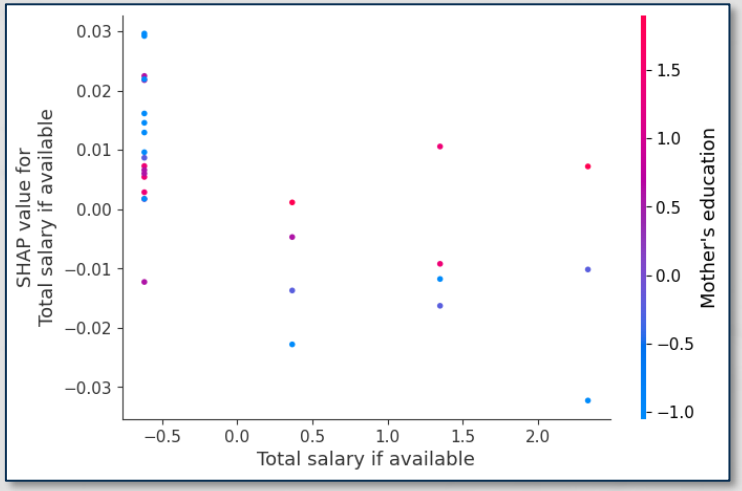

Figure 1: SHAP Dependence Analysis for Weekly Salary

#3) Social Indicators to Class Attendance

- Extracted personal and family attributes
- Calculated *Spearman* correlation

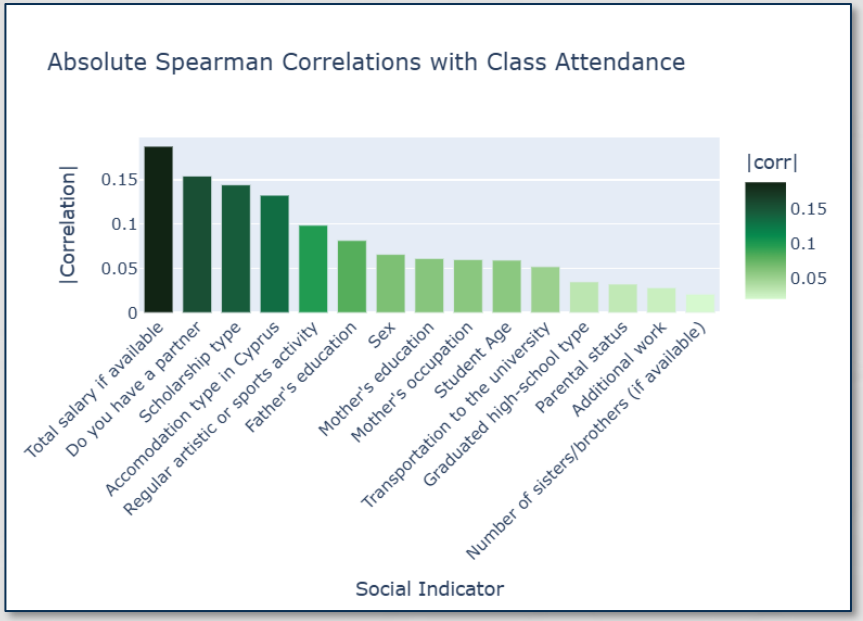

Figure 1: Spearman Correlations for Class Attendance

Conclusion & Limitations

- Data is small and self-reported
  - Possibility for noise
  - Non-generalizable
- Stronger and more generalizable analysis possible with larger dataset

Thank You!

Bibliography

- Yilmaz, N., & Şekeroğlu, B. (2019). *Higher Education Students Performance Evaluation* [Dataset]. UCI Machine Learning Repository. https://doi.org/10.24432/C51G82

  Annotation: This is the primary dataset used throughout the notebook (UCI ID 856), providing demographic, family, and educational-habit variables alongside outcomes like course grade and class attendance.
- Spearman, C. (1904). The proof and measurement of association between two things. *The American Journal of Psychology, 15*(1), 72–101.

  Annotation: Used to compute Spearman rank correlations when relating categorical-encoded social indicators to attendance; this is more robust than Pearson correlation when relationships are monotonic but not strictly linear.
- Mann, H. B., & Whitney, D. R. (1947). On a test of whether one of two random variables is stochastically larger than the other. *The Annals of Mathematical Statistics, 18*(1), 50–60.

  Annotation: Applied for non-parametric significance testing between cluster groups (Mann–Whitney U), avoiding normality assumptions and making it suitable for ordinal/categorical encodings.
- Ester, M., Kriegel, H.-P., Sander, J., & Xu, X. (1996). A density-based algorithm for discovering clusters in large spatial databases with noise. In *Proceedings of the Second International Conference on Knowledge Discovery and Data Mining (KDD'96)* (pp. 226–231).

  Annotation: Reference for DBSCAN, used as an alternative clustering approach that can label noise/outliers and does not require specifying the number of clusters in advance.
- Bergstra, J., & Bengio, Y. (2012). Random search for hyper-parameter optimization. *Journal of Machine Learning Research, 13*, 281–305.

  Annotation: Motivates the notebook's randomized hyperparameter search approach (via randomized search) as an efficient alternative to exhaustive grid search, especially when only a few hyperparameters strongly matter.
- Altmann, A., Toloşi, L., Sander, O., & Lengauer, T. (2010). Permutation importance: A corrected feature importance measure. *Bioinformatics, 26*(10), 1340–1347.

  Annotation: Supports the permutation importance procedure used to quantify how much shuffling each feature degrades model performance, producing a more model-agnostic view of importance than built-in impurity scores.
- Lundberg, S. M., & Lee, S.-I. (2017). A unified approach to interpreting model predictions. In *Advances in Neural Information Processing Systems* (NeurIPS).

  Annotation: Introduces SHAP values, which the notebook uses for post-hoc explainability (summary and dependence plots) to interpret how top features push predictions toward higher performance classes.